



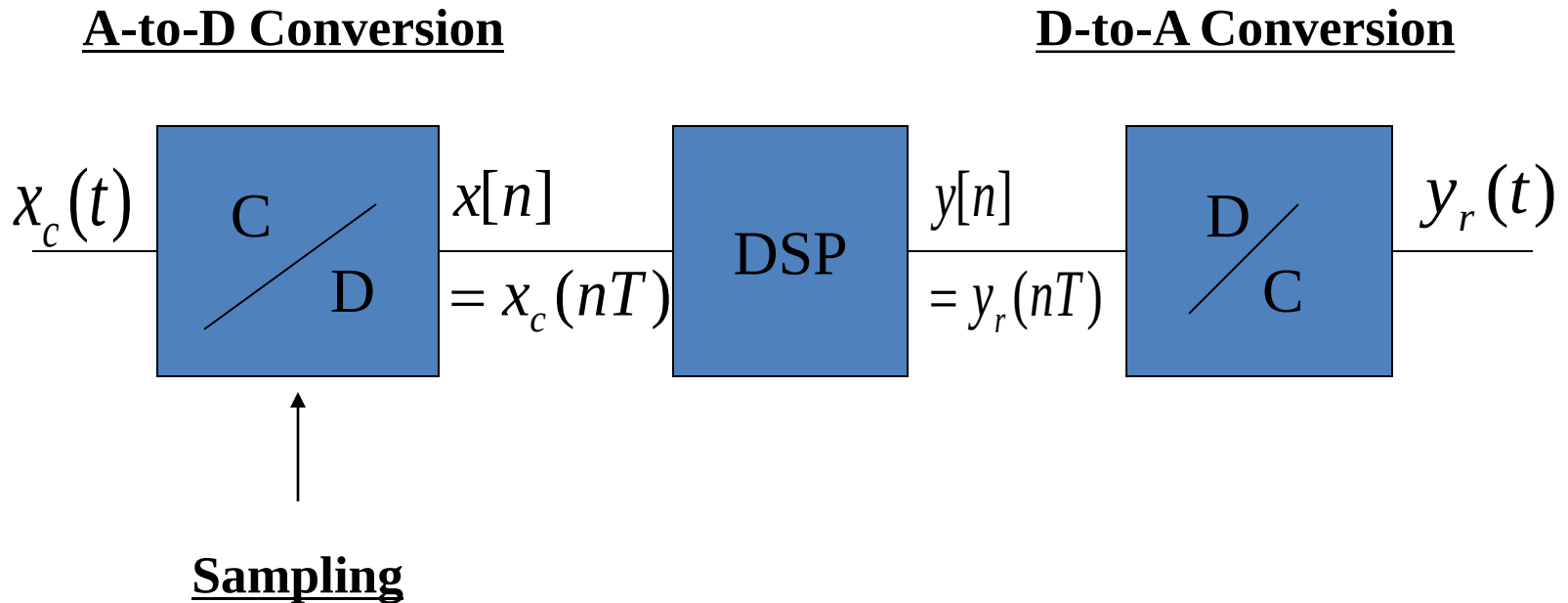
Advanced Digital Signal Processing

Lecture 8

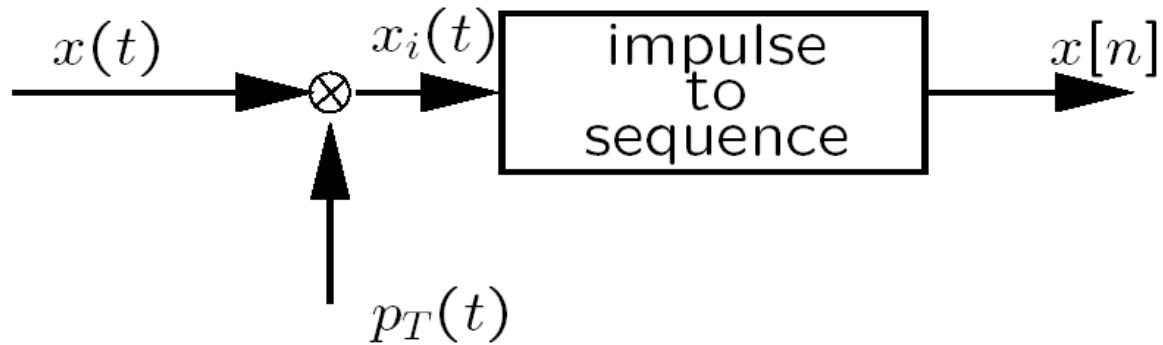
Sampling of Continuous-Time Signals

Dr. Tahir Zaidi

1. Analog-to-Digital Conversion and Sampling



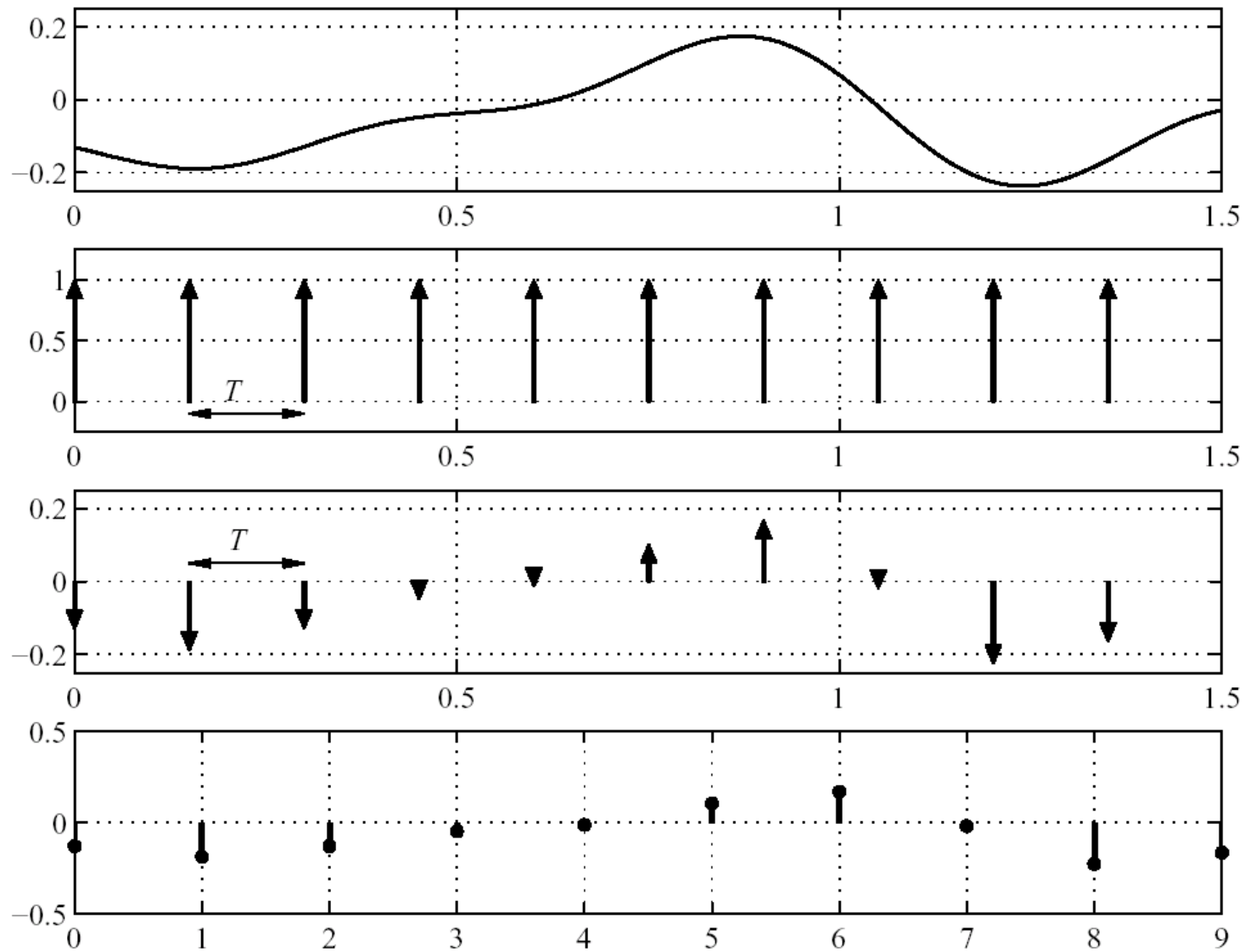
Periodic sampling



$$p_T(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT)$$

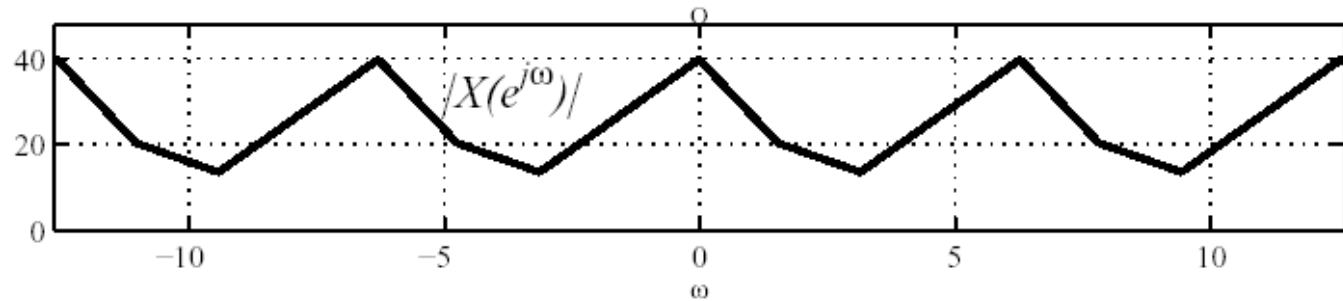
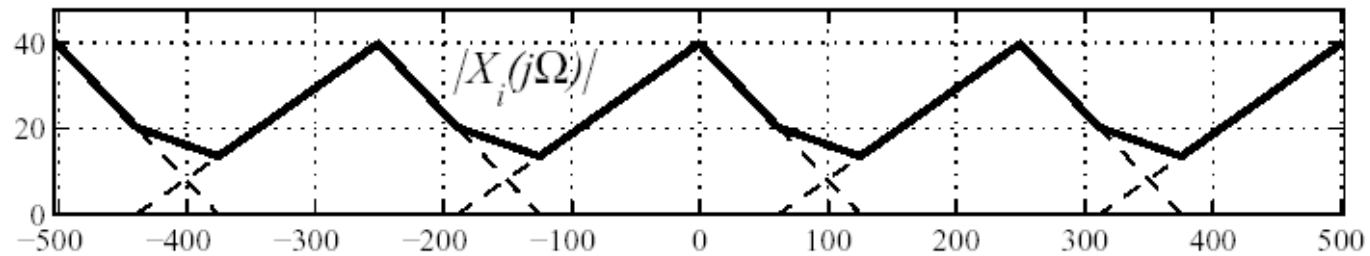
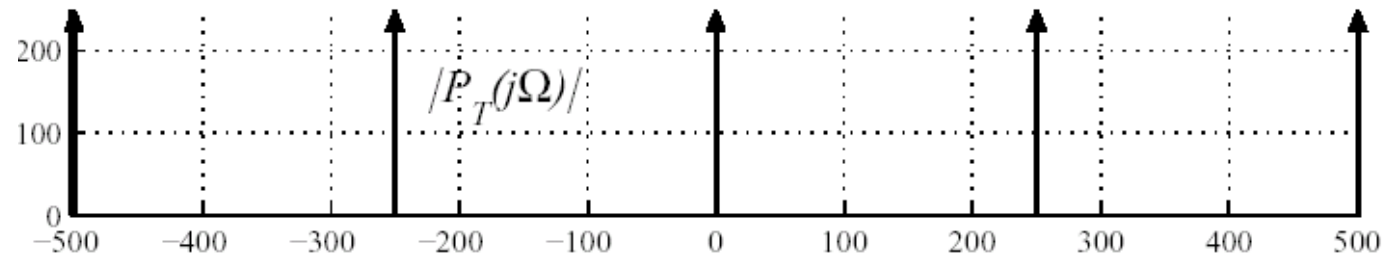
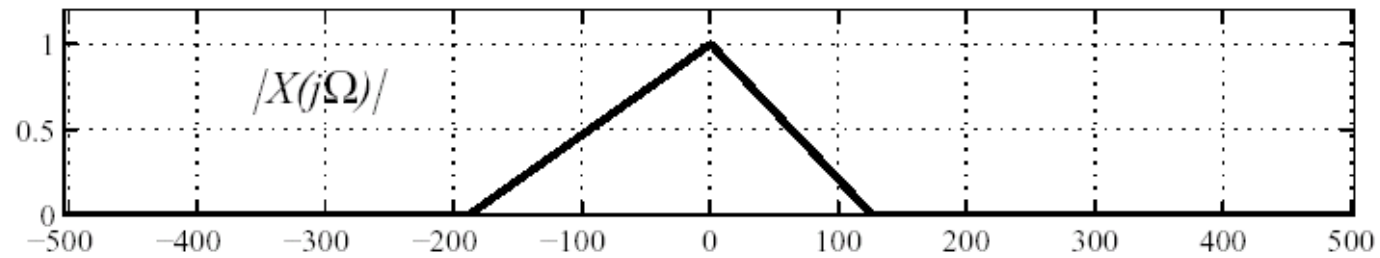
$$x_i(t) = x(t)p_T(t) = \sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT)$$

Periodic sampling: Time domain



Periodic sampling: Frequency domain

$$T=0.025101, \Omega_s=250.3211$$



Periodic sampling: Frequency domain

$$P_T(j\Omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{+\infty} \delta\left(\Omega - \frac{2\pi k}{T}\right)$$

$$X_i(j\Omega) = \frac{1}{2\pi} X(j\Omega) * P_T(j\Omega) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X\left(j\left(\Omega - \frac{2\pi k}{T}\right)\right)$$

$$x_i(t) = \sum_{n=-\infty}^{+\infty} x(nT) \delta(n - Nt) \xleftrightarrow{\mathcal{F}} X_i(j\Omega) = \sum_{n=-\infty}^{+\infty} x(nT) e^{-j\Omega nT}$$

$$x[n] = \xleftrightarrow{\mathcal{F}} X(e^{j\omega}) = \sum_{n=-\infty}^{+\infty} x[n] e^{-j\omega n},$$

with $x[n] = x(nT)$:

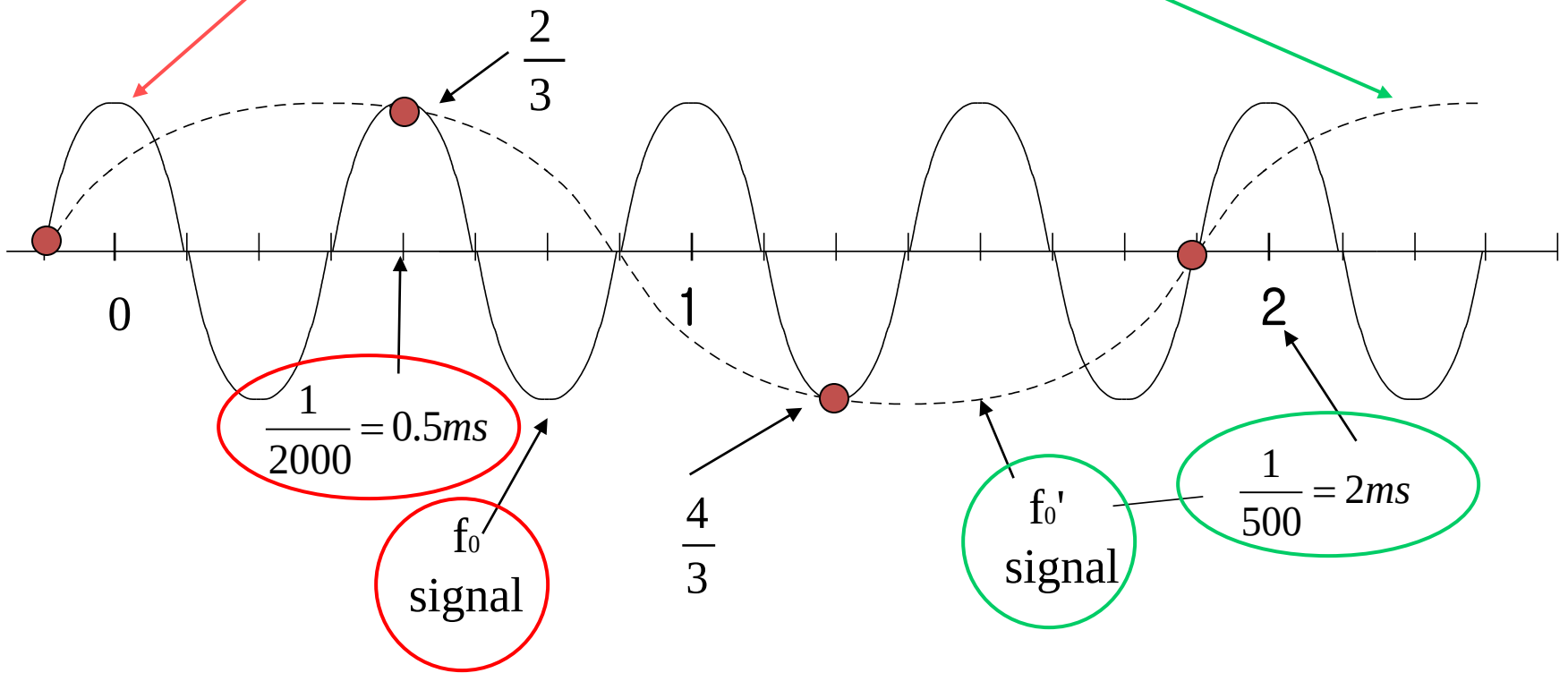
$$X(e^{j\omega}) = X_i(j\frac{\omega}{T}) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

Original signal

$$x_c(t) = \cos 4000\pi t$$

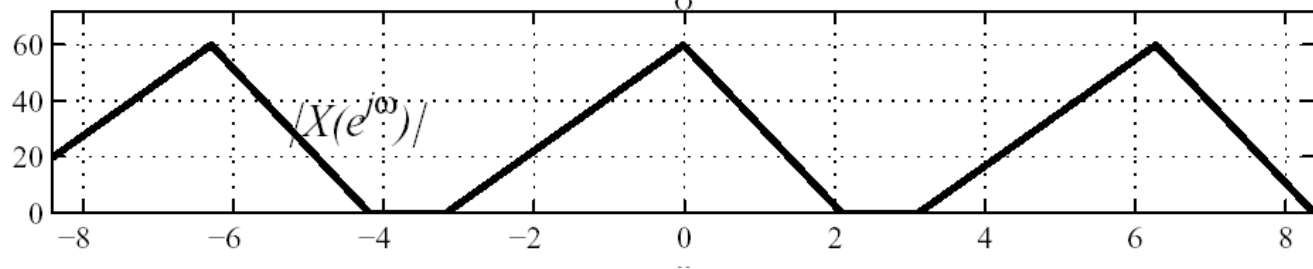
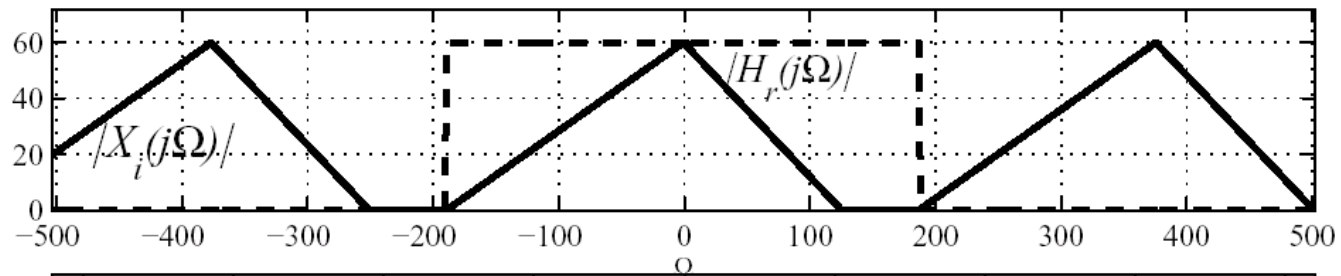
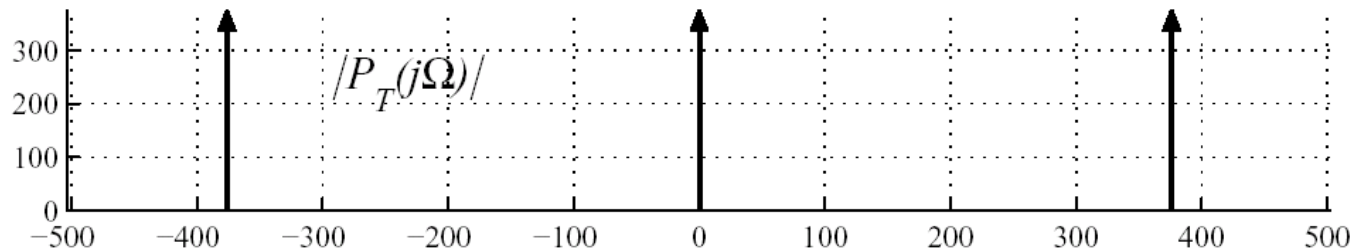
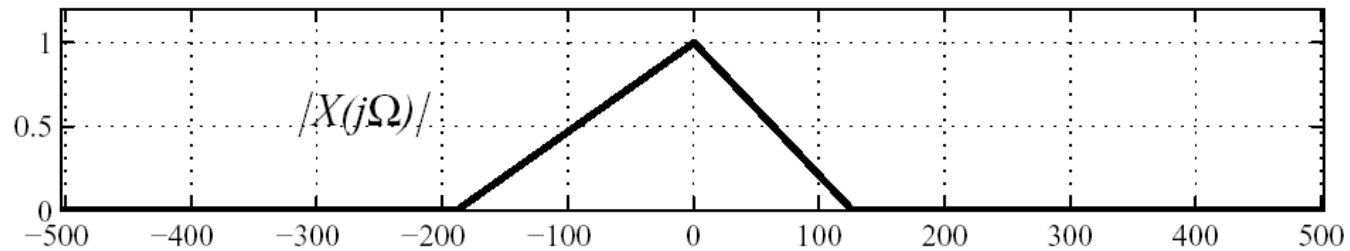
Aliased signal

$$x_c(t) = \cos 1000\pi t$$



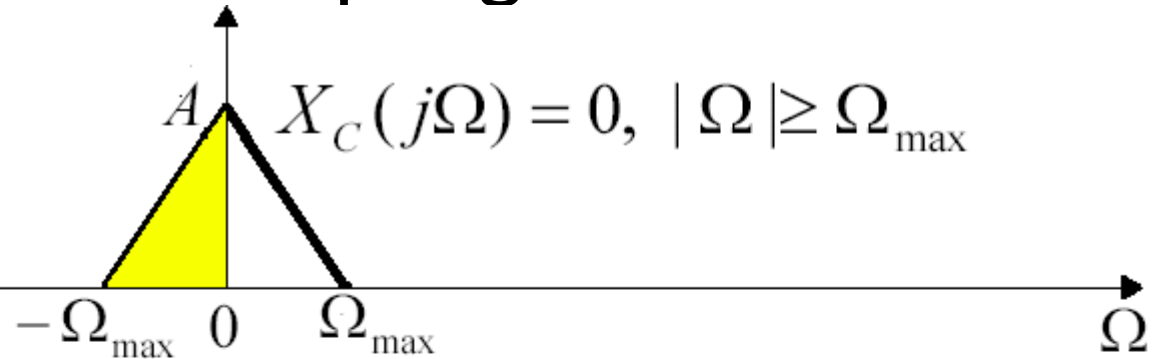
Nyquist Theorem

$$T=0.016706, \Omega_s=376.1106$$



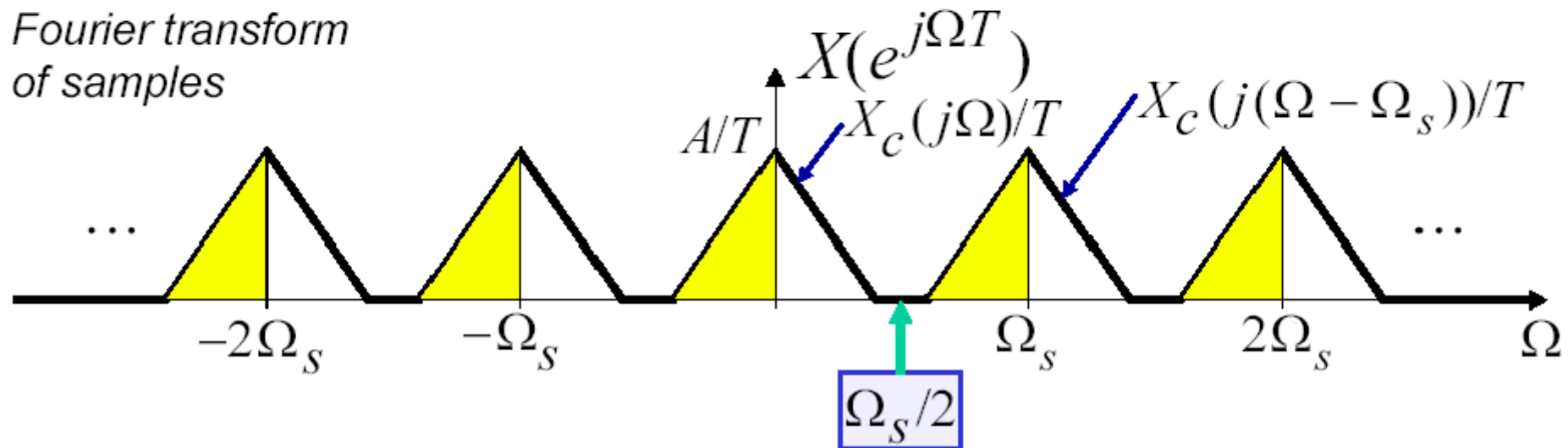
Oversampling

*"Typical"
bandlimited signal*



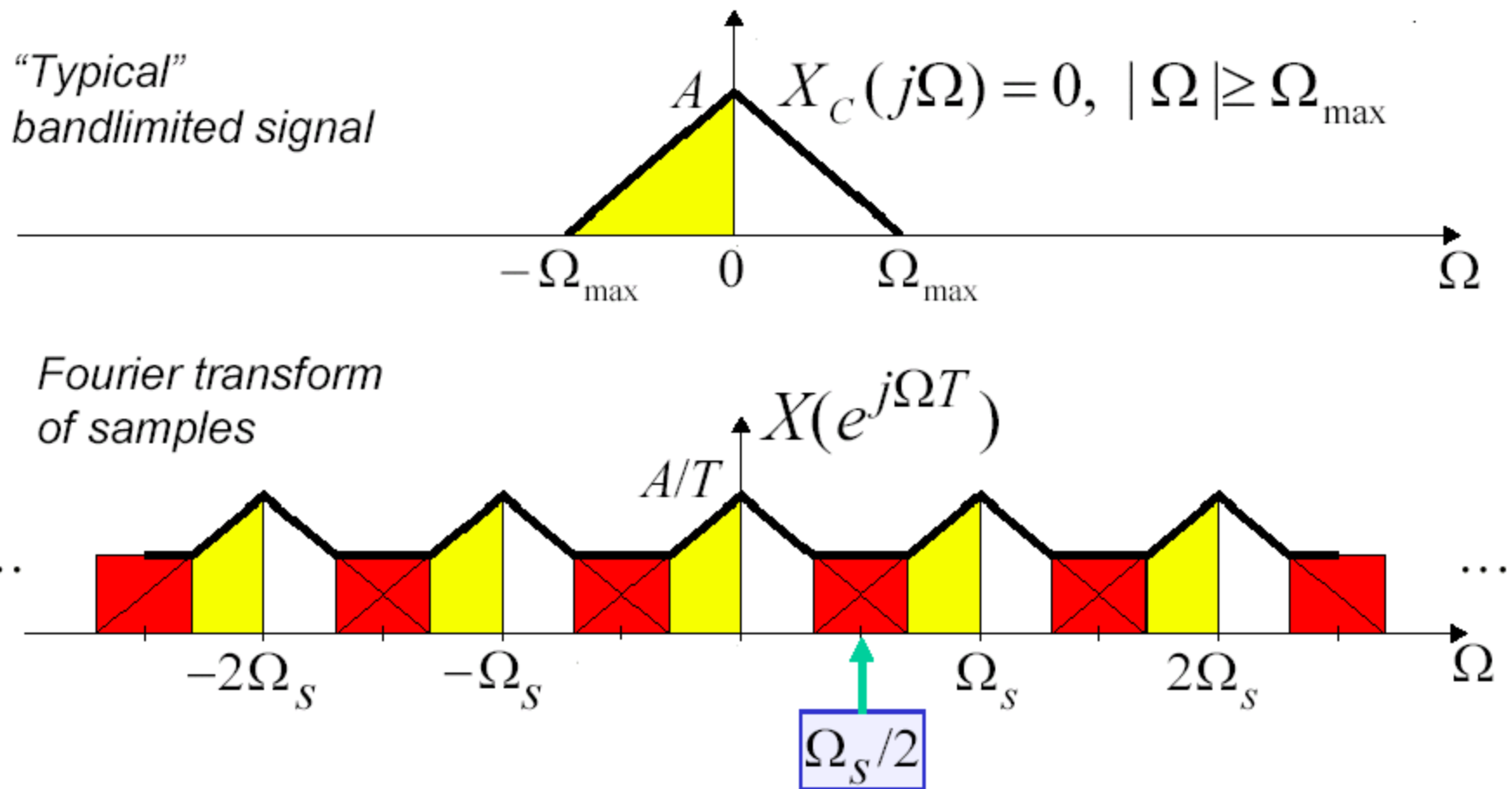
$$X(e^{j\Omega T}) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X_c(j(\Omega - k\Omega_s)) \quad \Omega_s = 2\pi/T$$

*Fourier transform
of samples*

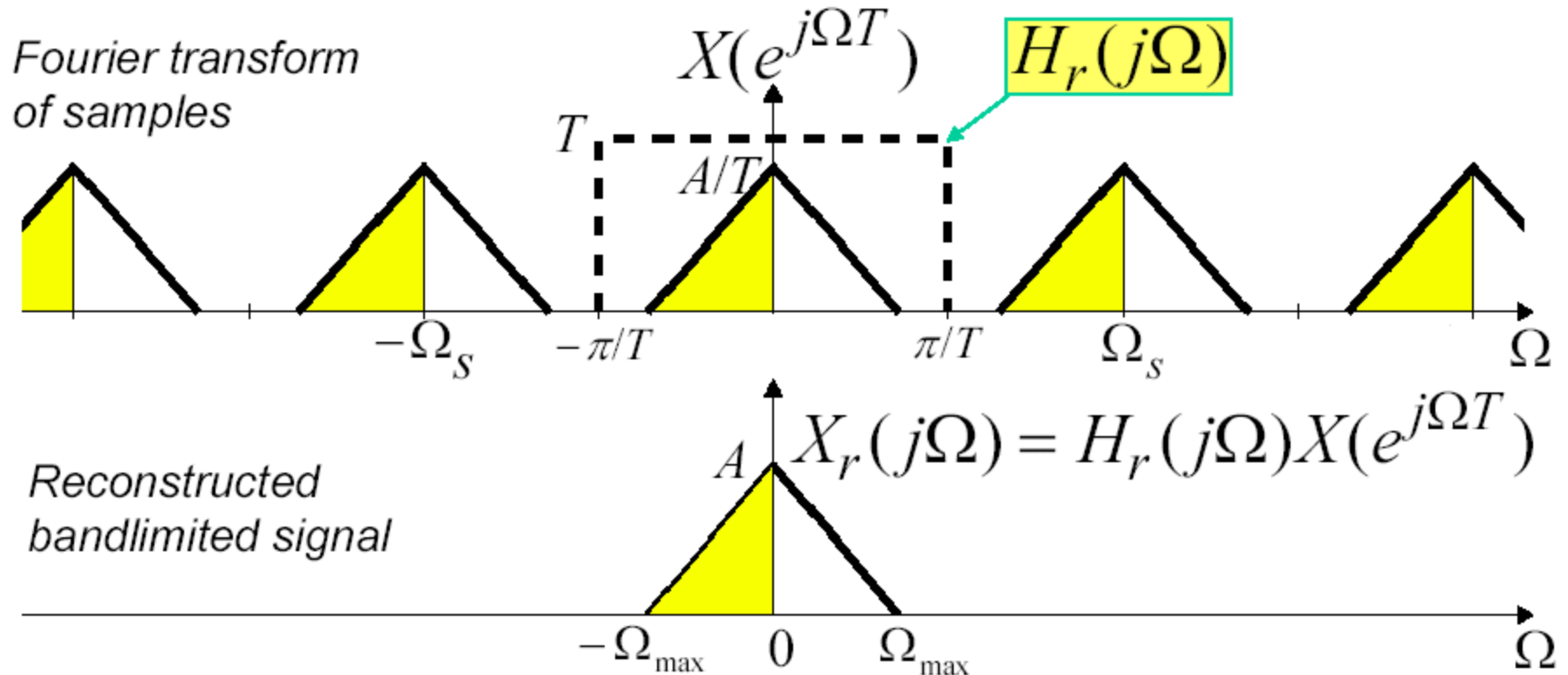


Undersampling (Aliasing Distortion)

If $\Omega_s < 2\Omega_{\max}$, the copies of $X_c(j\Omega)$ overlap, and we have **aliasing distortion**.

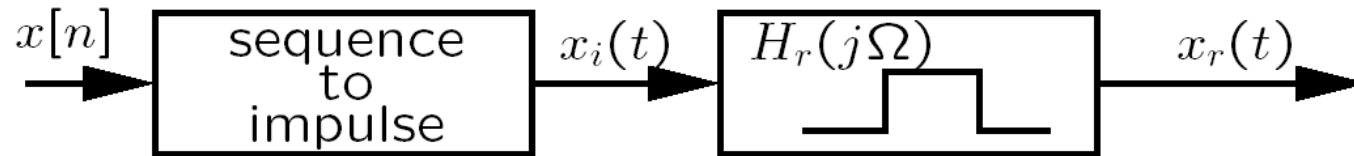


Reconstruction (Frequency Domain)



- If $\Omega_s > 2\Omega_{\max}$, the copies of $X_c(j\Omega)$ do not overlap, so $X_r(j\Omega) = H_r(j\Omega)X(e^{j\Omega T}) = X_c(j\Omega)$, and we get perfect reconstruction.

Nyquist Theorem



Nyquist theorem: Let $x(t)$ be band-limited so that $X(j\Omega) = 0$ for $|\Omega| \geq \Omega_N$. Then $x(t)$ can be perfectly recovered from its samples $x(nT)$ taken with any period T such that

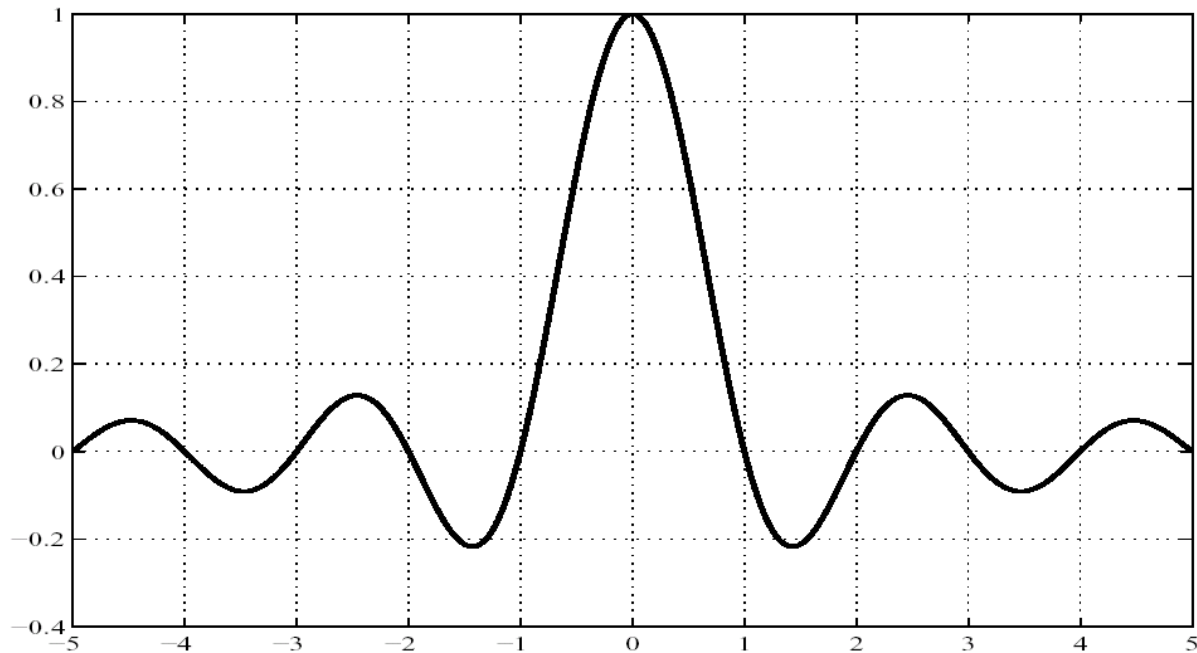
$$\Omega_s = \frac{2\pi}{T} \geq 2\Omega_N$$

Band-limited interpolation

$$X_r(j\Omega) = X_i(j\Omega)H_r(j\Omega)$$

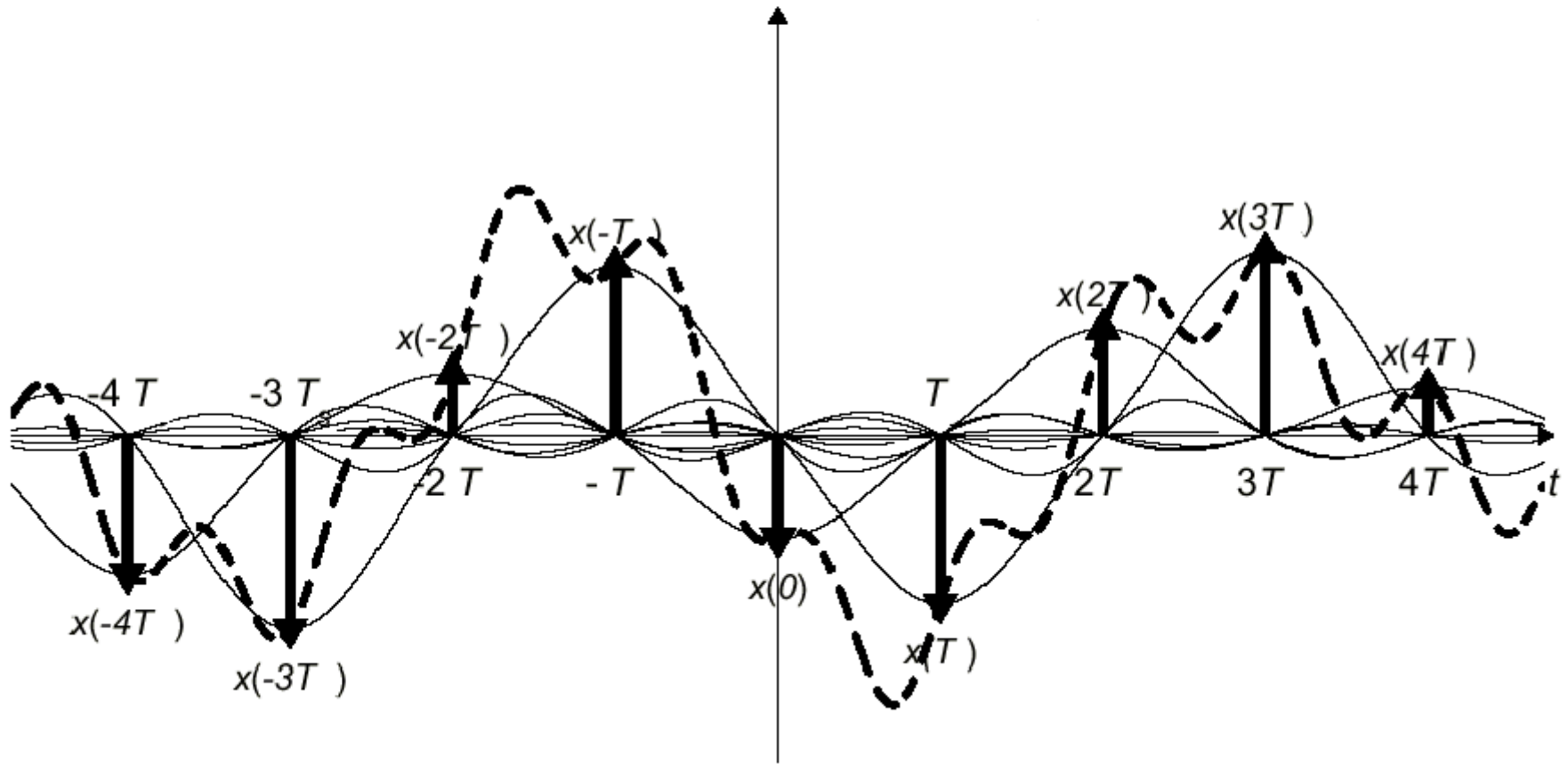
$$x_r(t) = \sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT) * h_r(t)$$

$$= \sum_{n=-\infty}^{+\infty} x(nT) \frac{\sin \pi(t - nT)/T}{\pi(t - nT)/T}$$

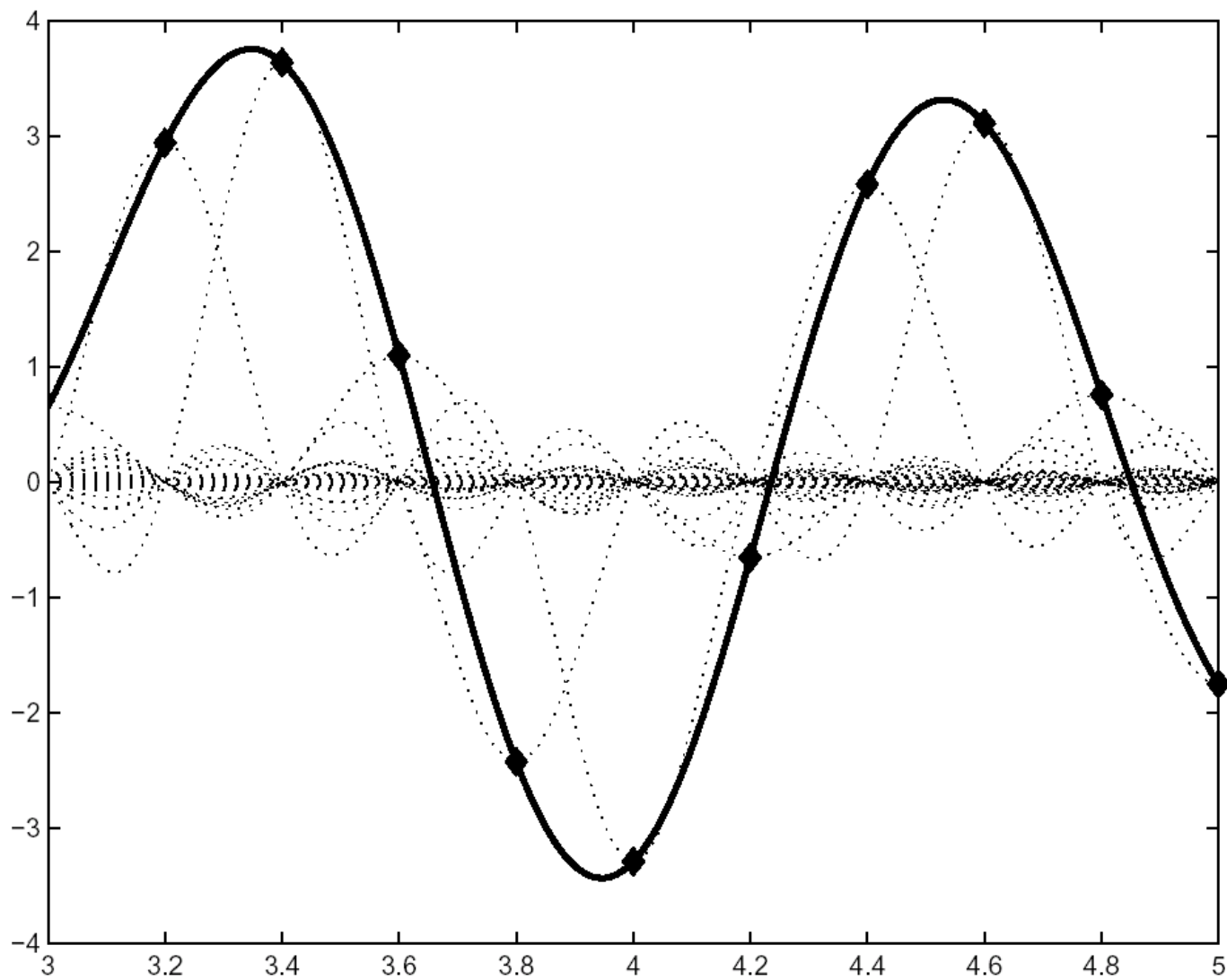


Reconstruction Time Domain

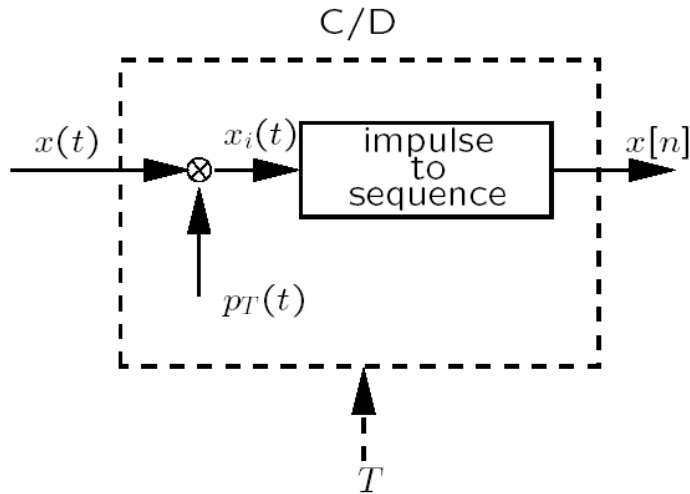
$$x_r(t) = \sum_{n=-\infty}^{\infty} x(nT)h_r(t-nT) = \sum_{n=-\infty}^{\infty} x(nT)\frac{\sin[\pi(t-nT)/T]}{\pi(t-nT)/T}$$



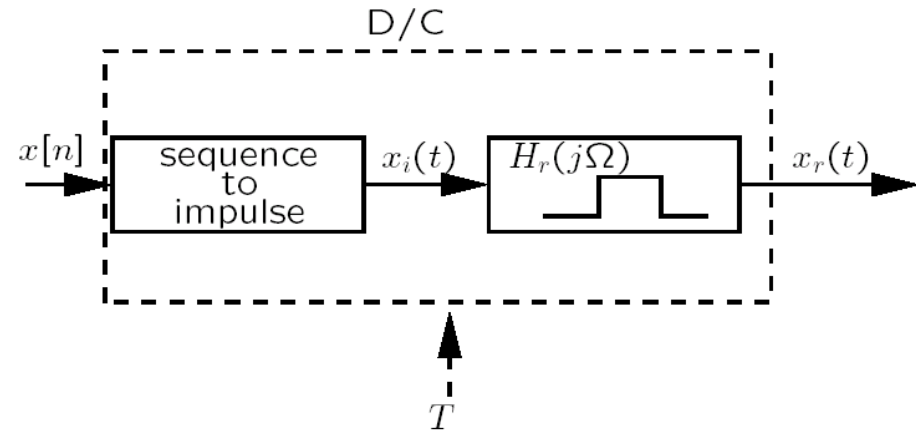
Band-limited interpolation



Ideal D/C and C/D



$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X\left(j\left(\frac{\omega}{T} - \frac{2\pi k}{T}\right)\right)$$

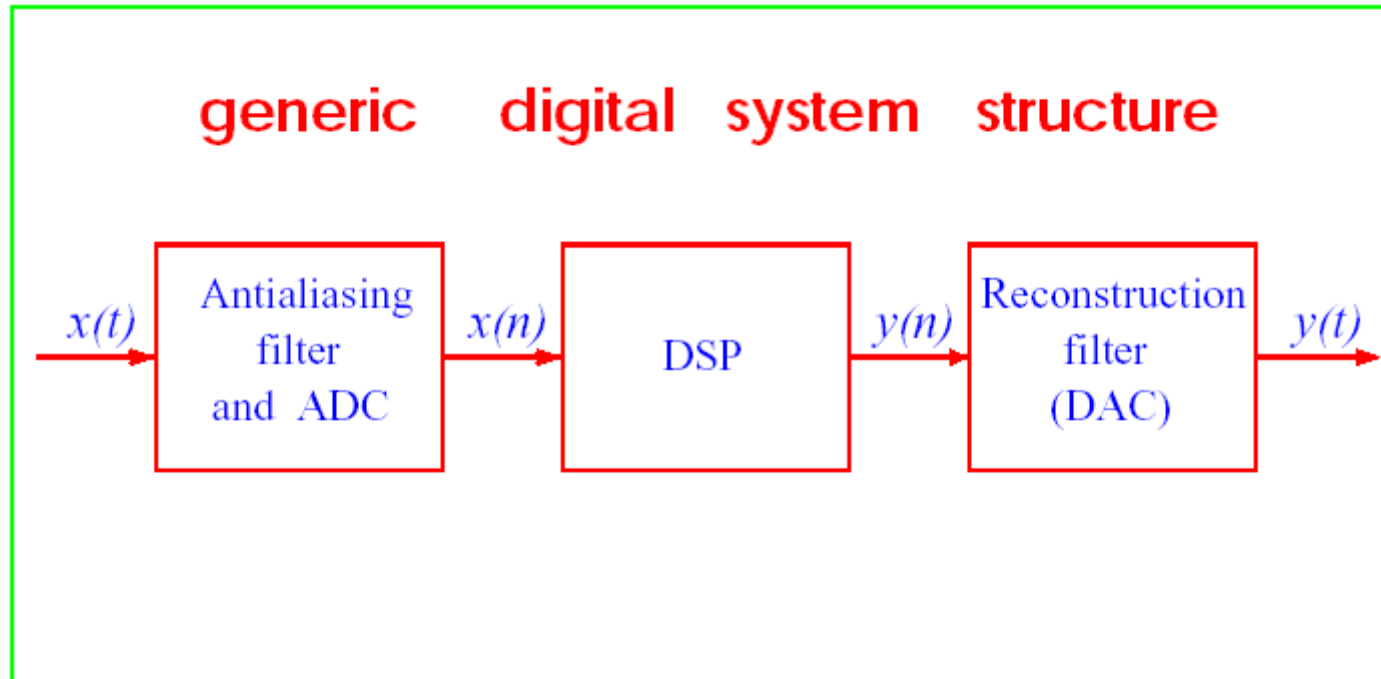


$$X_r(j\Omega) = T X(e^{j\Omega T})$$

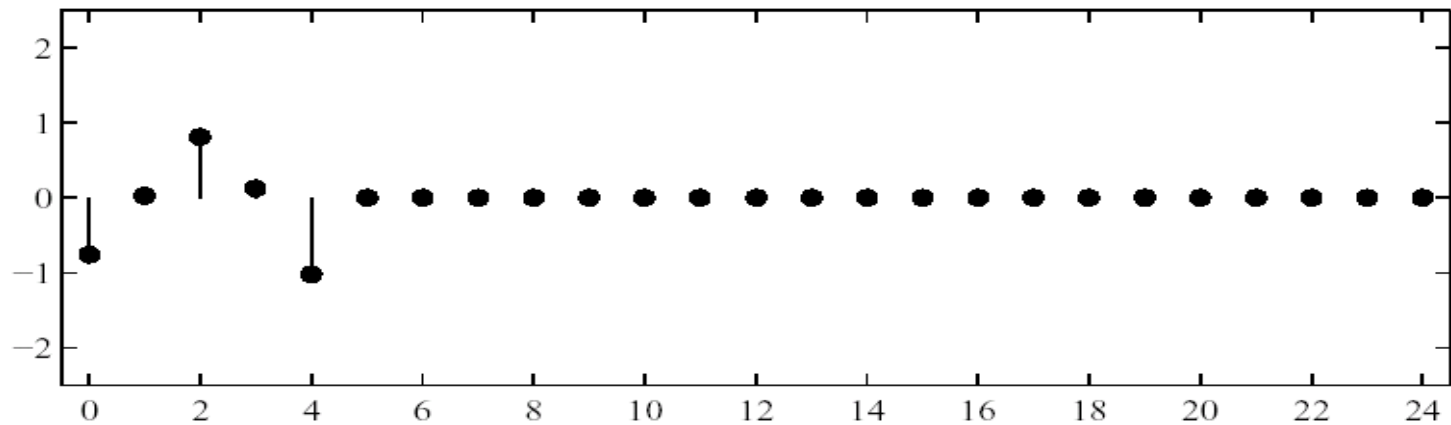
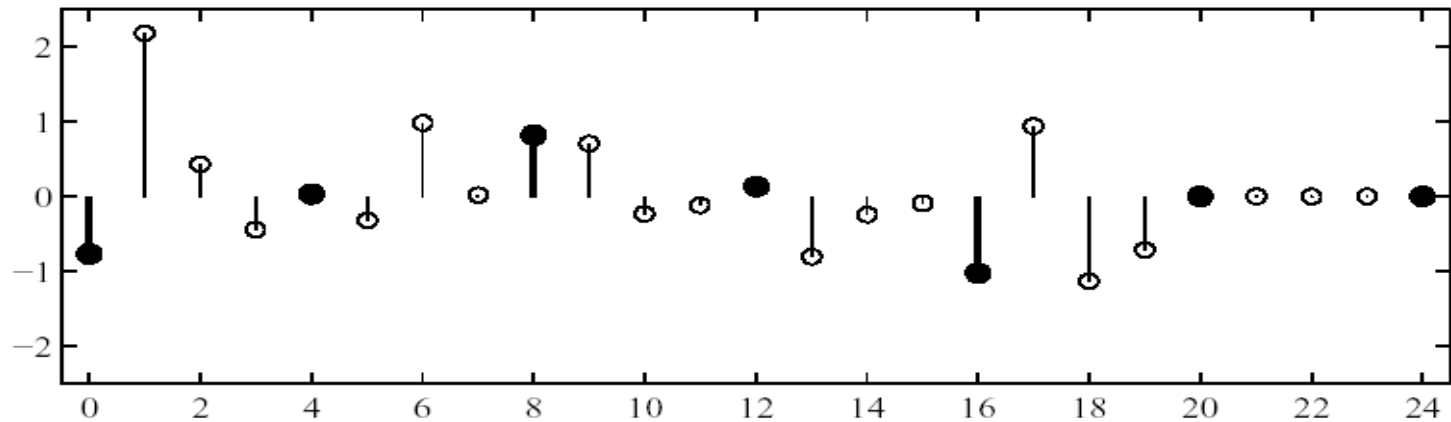
If T is small enough (Nyquist), then

$$X_r(j\Omega) = X(j\Omega)$$

DT processing of continuous time signals



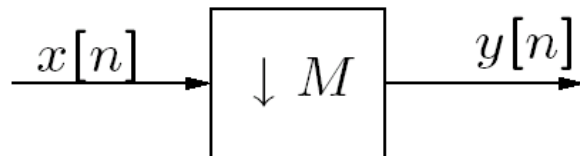
Downsampling in time-domain



decim



Downsampling

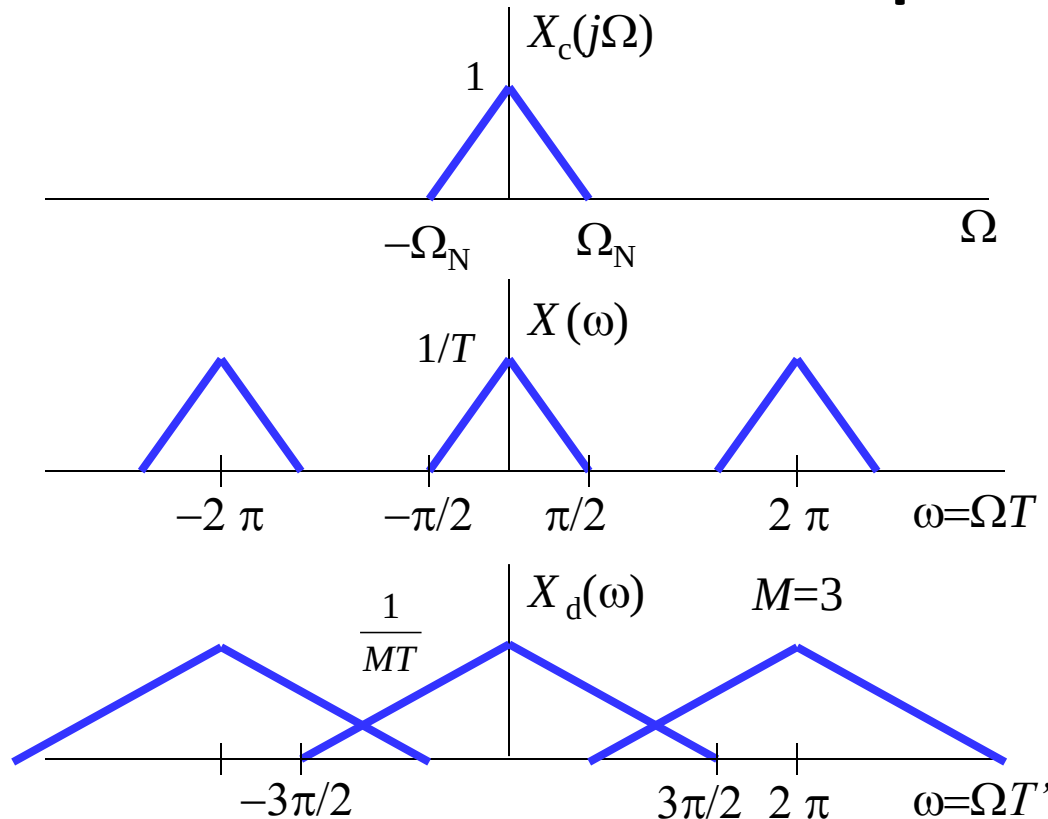


$$y[n] = x[Mn]$$

$$X(e^{j\omega}) = \frac{1}{T} \sum_{k=-\infty}^{+\infty} X \left(j \left(\frac{\omega - 2\pi k}{T} \right) \right)$$

$$\begin{aligned} Y(e^{j\omega}) &= \frac{1}{MT} \sum_{k=-\infty}^{+\infty} X \left(j \left(\frac{\omega - 2\pi k}{MT} \right) \right) \\ &= \frac{1}{M} \sum_{m=0}^{M-1} \frac{1}{T} \sum_{n=-\infty}^{+\infty} X \left(j \left(\frac{\omega - 2\pi m - 2\pi Mn}{MT} \right) \right) \\ &= \frac{1}{M} \sum_{m=0}^{M-1} X(e^{j(\frac{\omega - 2\pi m}{M})}) \end{aligned}$$

Downsampling

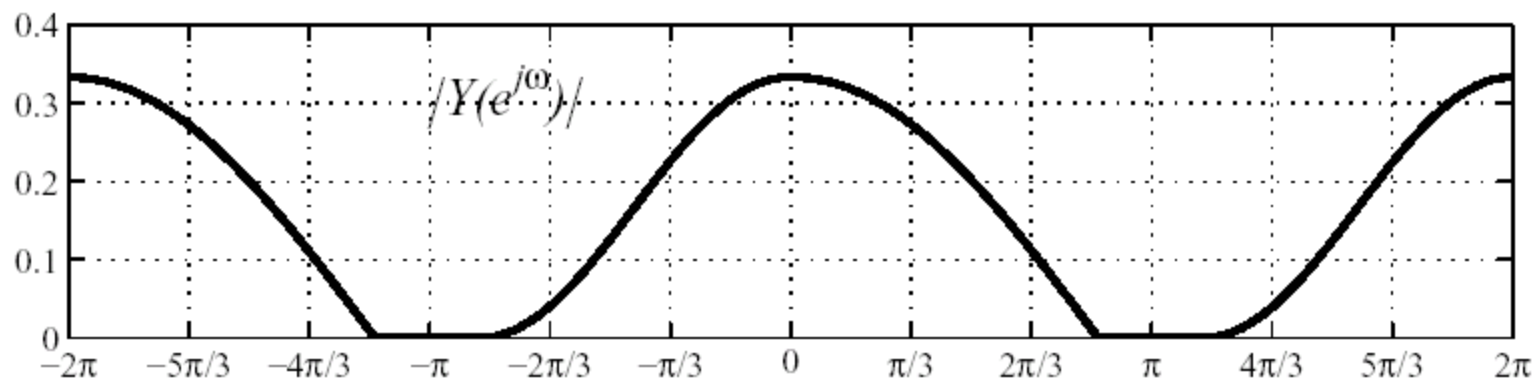
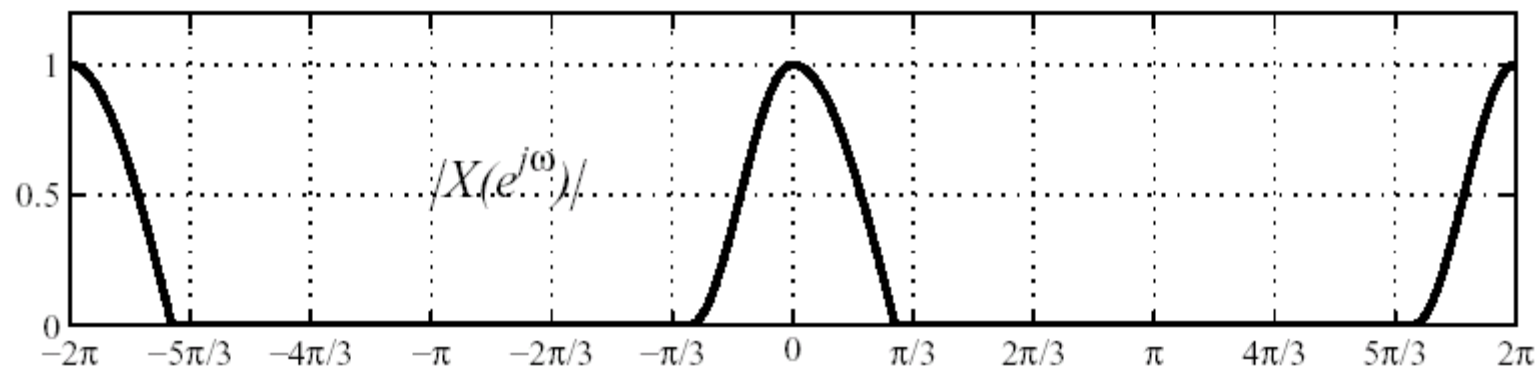


Sample the analog bandlimited signal every T time units

Downsampling by M generates baseband plus $M-1$ copies of baseband per period of frequency domain

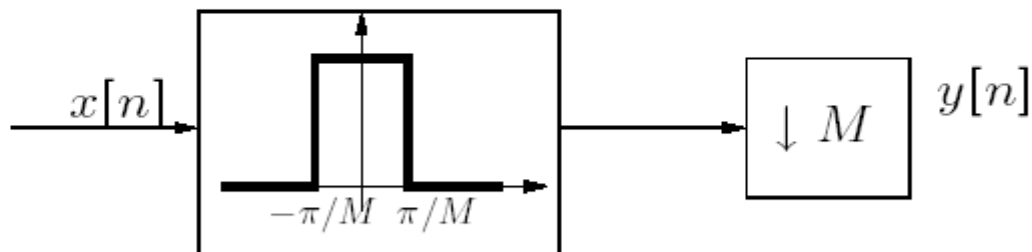
Aliasing occurs: avoid aliasing by pre-filtering with lowpass filter with gain of 1 and cutoff of π/M to extract baseband

Downsampling in Frequency-domain



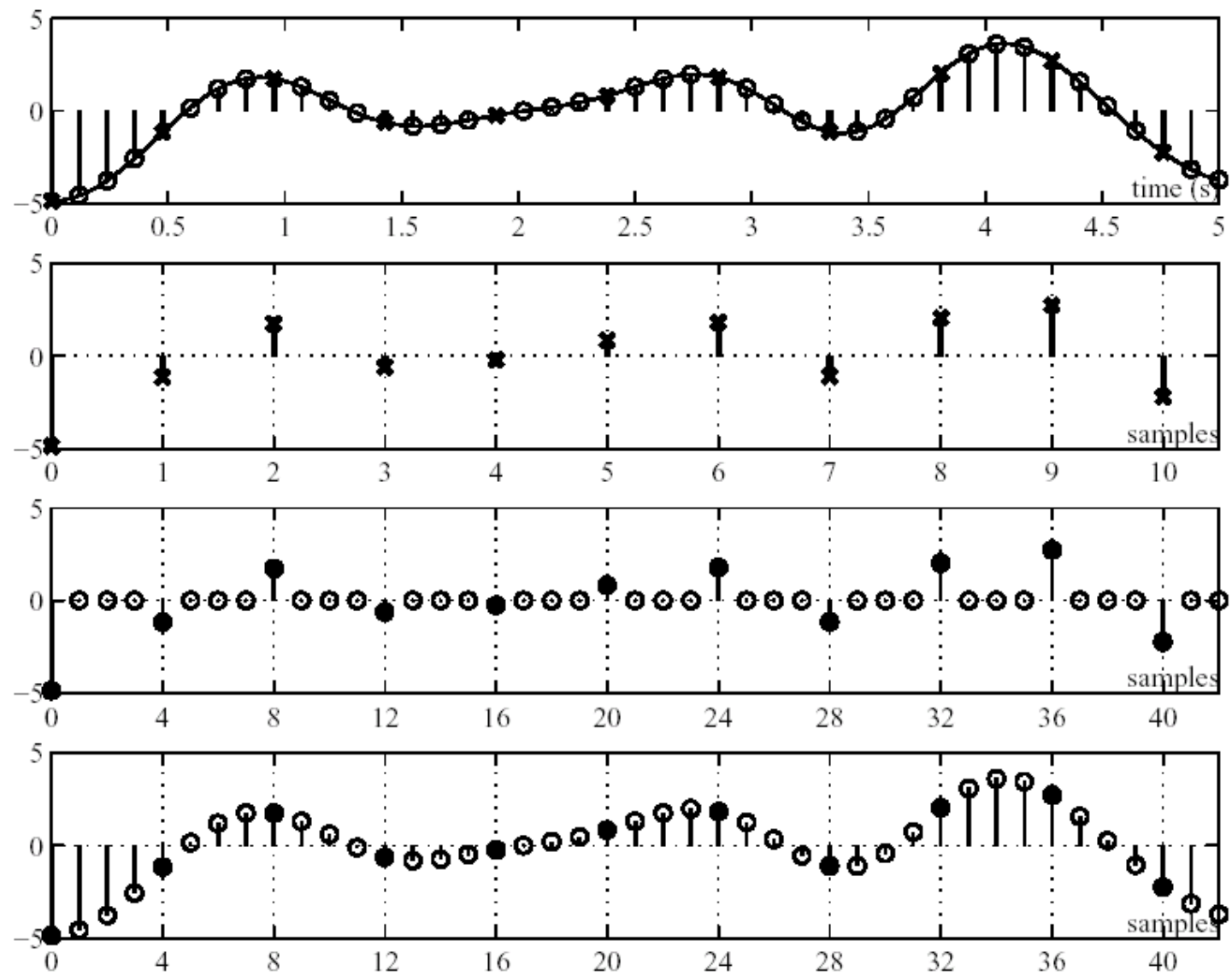
Aliasing and pre-filtering

- No aliasing if $X(e^{j\omega}) = 0$ for $\frac{\pi}{M} \leq \omega \leq \pi$
- If not satisfied: pre-filtering is needed:

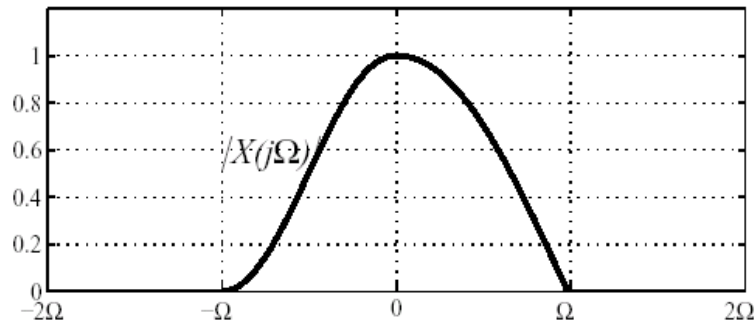
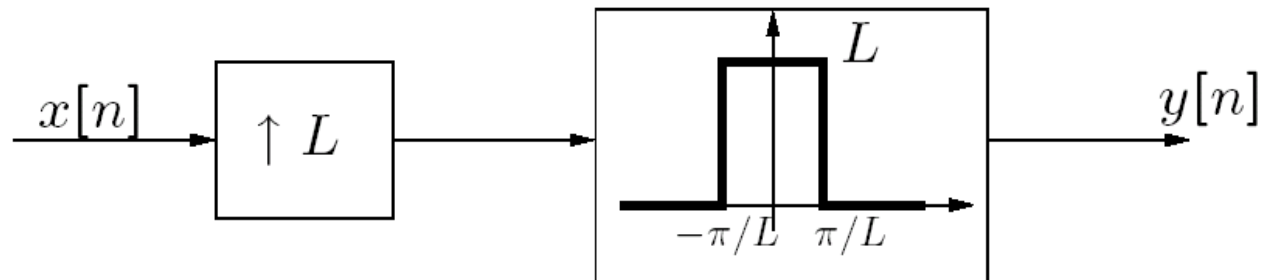


Note: in this case, $y[n] \neq$ samples of $x(t)$!!

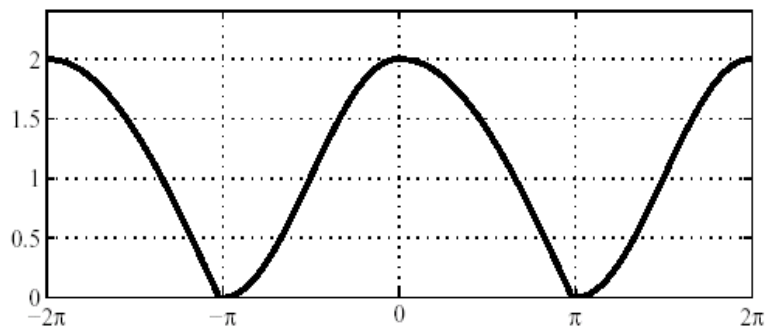
Upsampling



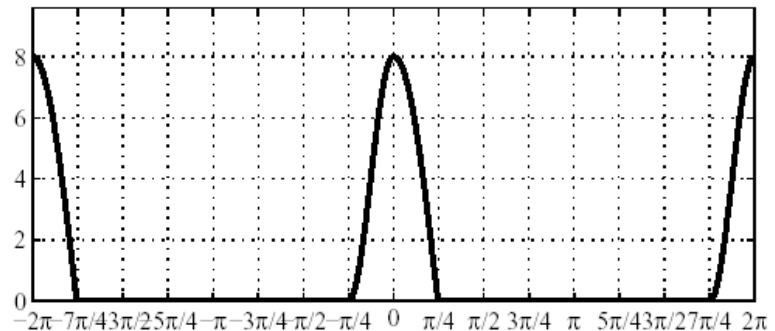
Upsampling



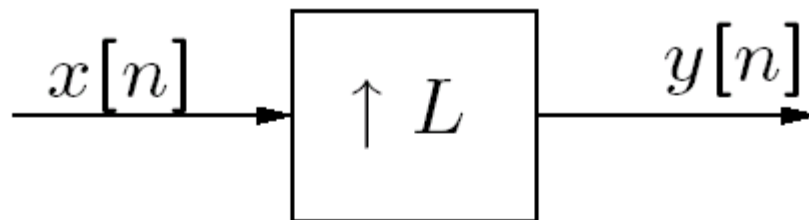
$$\frac{2\pi}{T} = 2\Omega_N$$



$$T' = \frac{T}{L}, \frac{2\pi}{T'} = 2L\Omega_N$$



Aliasing and pre-filtering

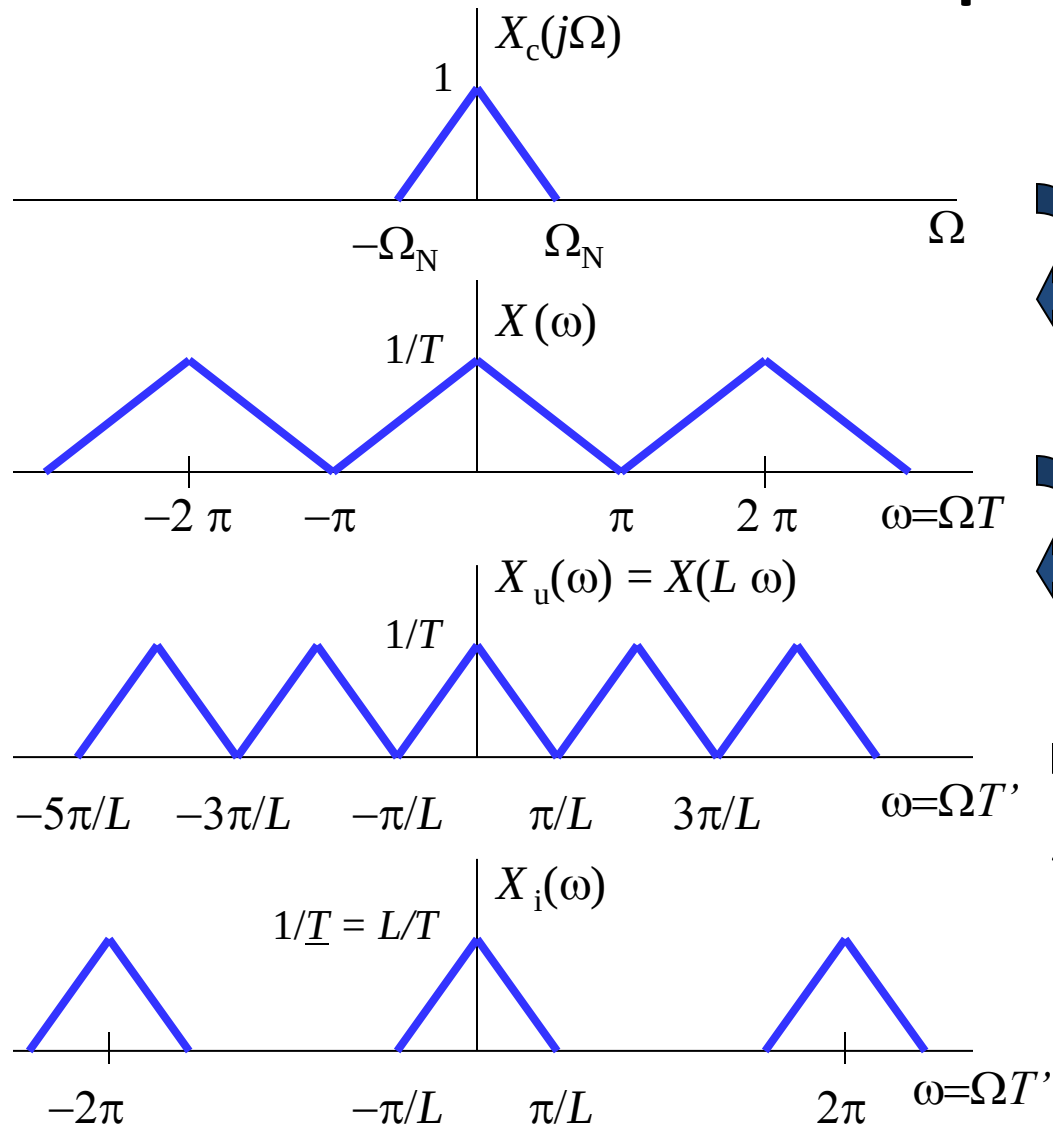


$$y[n] = \begin{cases} x[n/L] & n = kL \\ 0 & \text{otherwise} \end{cases}$$

$$Y(e^{j\omega}) = X(e^{j\omega L})$$

$$Y(z) = X(z^L)$$

One-Dimensional Upsampling



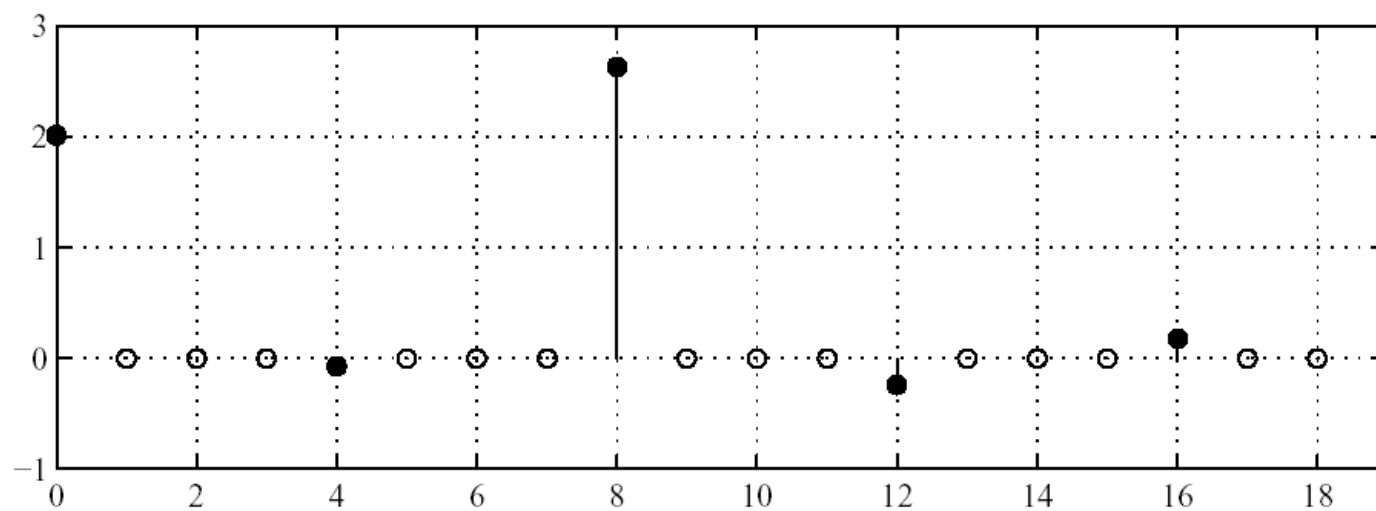
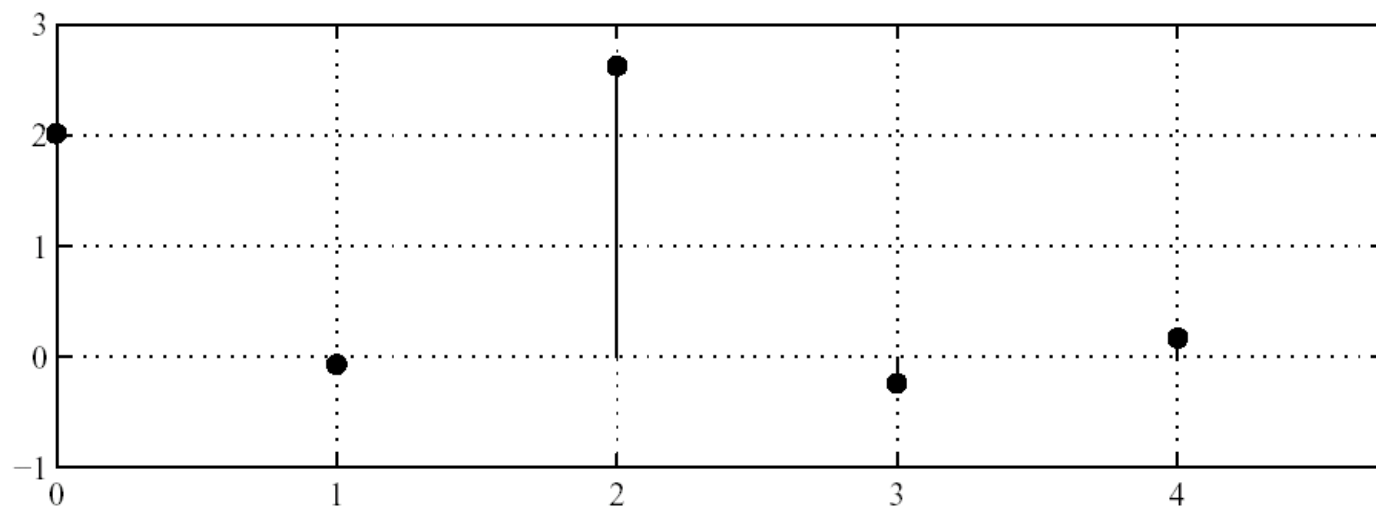
Sample the analog bandlimited signal every T time units

Upsampling by L gives L images of baseband per 2π period of ω

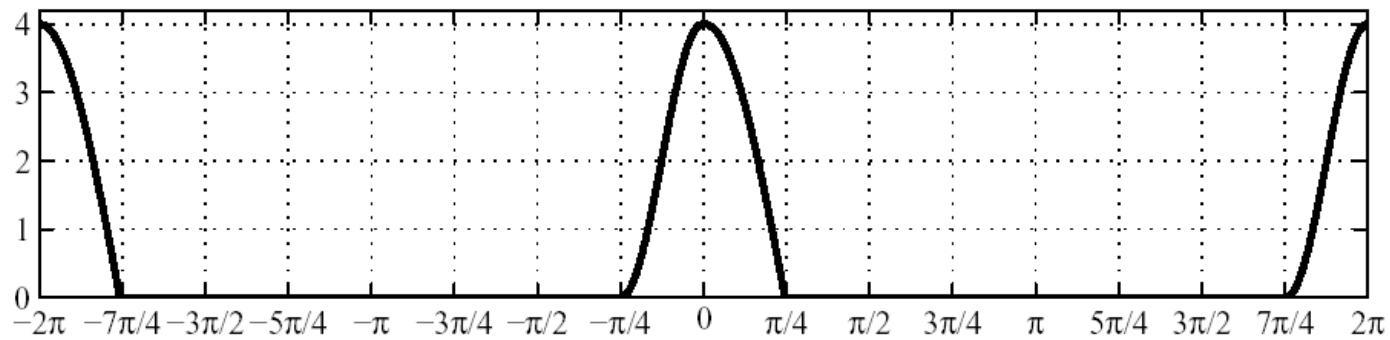
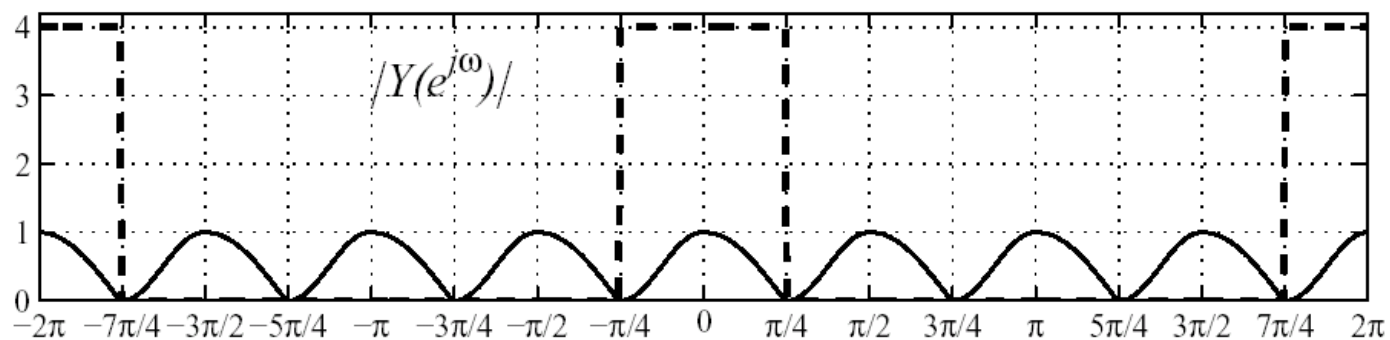
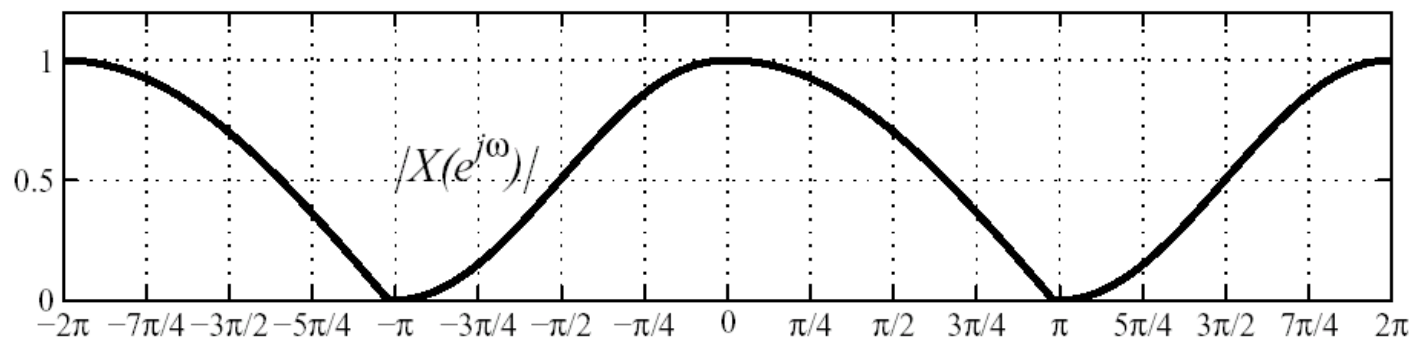
Apply lowpass interpolation filter with gain of L and cutoff of π/L to extract baseband

Fig. 3.22 Oppenheim & Schaffer, 1989.

Zero insertion



Zero insertion



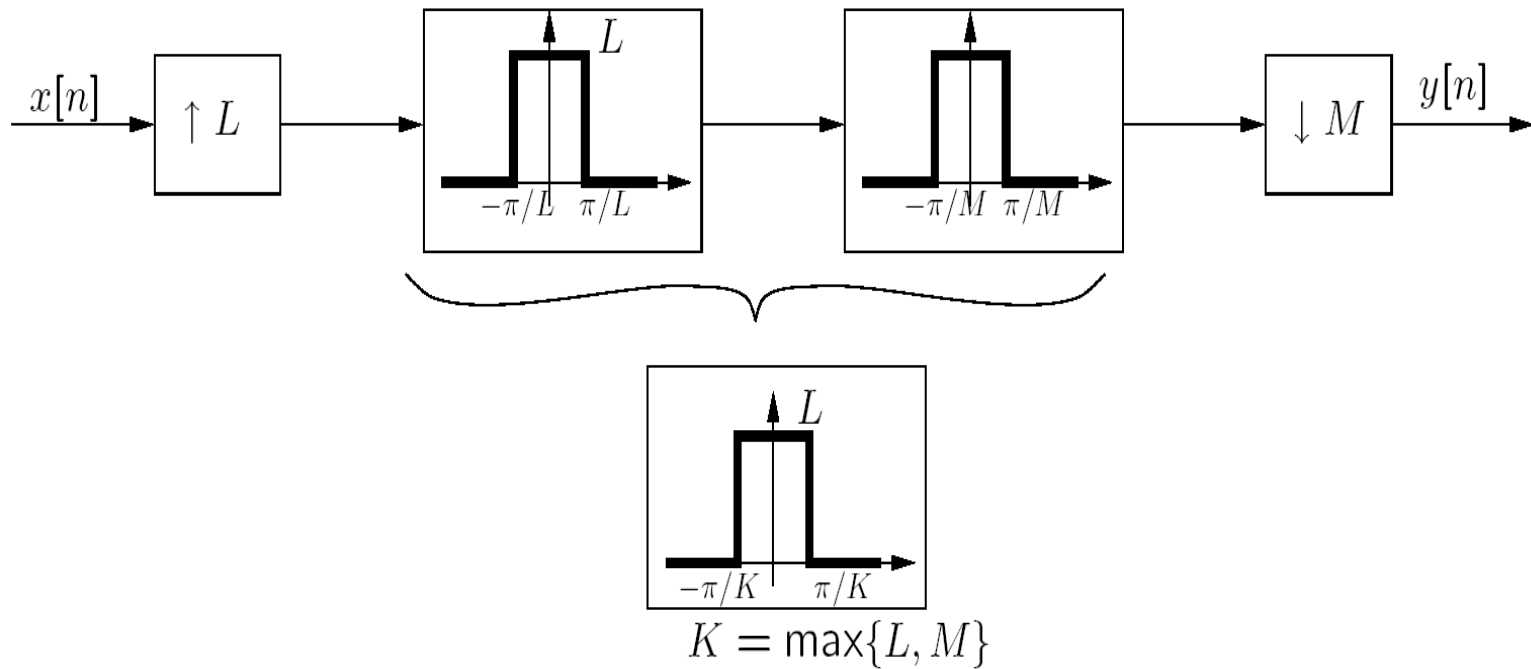
Zero insertion

- Ideal low-pass filter approximated
- Or replaced by linear interpolation:

$$h_{\text{lin}}[n] = \begin{cases} 1 - \frac{|n|}{L} & |n| \leq L \\ 0 & \text{elsewhere} \end{cases}$$

Zero insertion

- Cascade up- and down-sampling:



Zero insertion

$$L = 3, M = 2$$

